

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Computer Applications (MCA) /Master of Computer Applications (MCA) Lateral
Semester - IV / Semester - II
University Examination May 2018
CA836 - Frameworks and Applications

Date: 21.05.18, Monday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Answer to the two sections must be written in the separate answer books.
2. Answer to different questions should start on new page.
3. Figure to the right indicate full marks.
4. Make suitable assumptions wherever necessary and mention them clearly.

SECTION – I**Q-1 Do as directed.****[A] State whether the following statements are [True/False].****[05]**

1. Default Scope of any bean within the container is *Singleton*.
2. *Set* allows us to inject duplicate values.
3. Correct full form of OXM is Object XML Model.
4. *Dependency Injection* (DI) makes the programming code tightly coupled.
5. It is compulsory to specify “id” attribute of *inner bean*.

[B] Fill in the blanks.**[06]**

1. _____ is used to get/read the messages from the *property file*.
a) getProperty() b) getMessage() c) getKey() d) get()
2. _____ is set of theories which is widely accepted enough to serve as the guiding principal of research.
a) Software Framework b) Conceptual Framework c) Both d) None of these
3. _____ is limitation of *autowiring*.
a) Overriding possibility b) AutoWire Primitive data types c) Confusing nature d) All of these
4. _____ is implementation of *ApplicationContext*.
a) ClassPathXmlApplica b) FileSystemXmlApplica c) Both d) None of these
tionContext tionContext
5. _____ can publish events to beans that are registered as listeners.
a) ApplicationContext b) BeanFactory c) Both d) None of these
6. Return type of *executeQuery()* is _____.
a) int b) ResultSet c) Query d) boolean

Q-2 Answer the following questions. [Any Three]**[12]**

- [A]** Draw the detailed *Spring Framework Architecture*. Explain any two modules of it.
- [B]** Explain *Inner Bean* with proper example.
- [C]** What is *Auto Wiring* in Spring? List and explain any two types of it with proper example.
- [D]** Explain the concept of *Setter Injection* with proper example.
- [E]** What is the use of *Collection classes* in Spring? Explain *Map* collection with proper example.

Q-3 Answer the following questions. [Any Three]**[12]**

- [A]** Write a note on *Factory Design Pattern*.
- [B]** Explain *Object Referencing* of a Bean with proper example.
- [C]** Explain the concept of *Constructor Injection* with proper example.
- [D]** Write a note on *ResourceBundleMessageSource* class and explain it with proper example.
- [E]** What is *IoC container*? What are the types of *IoC containers*? Explain them.

SECTION – II

Q-4 Do as directed

[A] State whether the following statements are [True/False]. [05]

1. *@Aspect* annotation is used to indicate a class as an *Aspect class*.
2. OOP provides modularity by *Class* whereas AOP provides modularity by *Object*.
3. *execute()* method is used to fire *create* query in Hibernate.
4. *SessionFactory* object is used to create the object of *Transaction*.
5. *ORM* is a programming technique that maps the *object* to the *data* stored in the database.

[B] Fill in the blanks. [06]

1. _____ annotation is used to provide *auto increment* to a particular field.
a) *@AutoIncrement* b) *@Id* c) *@GeneratedValue* d) *@Unique*
2. In Spring Web MVC, *DispatcherServlet* class works as _____.
a) *WebContainer* b) *Front Controller* c) *Model* d) *View*
3. The job/purpose of an *Aspect* is called _____.
a) *Aspect* b) *PointCut* c) *Advice* d) *None of these*
4. *Where* clause is defined by _____ terminology of *AOP framework*.
a) *Advice* b) *JoinPoint* c) *PointCut* d) *None of these*
5. _____ is used to provide *filtration* in hibernate.
a) *ProjectionList* b) *Projections* c) *Selection* d) *Restrictions*
6. *before()* method is required to override if the class implements _____ interface.
a) *MethodInterceptor* b) *ThrowsAdvice* c) *MethodBeforeAdvice* d) *None of these*

Q-5 Answer the following questions. [Any Three] [12]

- [A] Describe *Hibernate Configuration File*.
[B] Draw and explain *Spring Web MVC Architecture*.
[C] Define the following annotation used in *Hibernate*:
(i) *@Entity* (ii) *@Id* (iii) *@OneToOne* (iv) *@Transient*
[D] Explain *Advice*, *JoinPoint* and *PointCut* with its diagram.
[E] Write a *code snippet* to execute below query using *Criteria* in *Hibernate*:
Select Empid, Empname From Emp Where Empcity = 'Anand'

Q-6 Answer the following questions. [Any Three] [12]

- [A] Draw *Hibernate Architecture* and explain any two component of it.
[B] Describe *DispatcherServlet.xml* file used in *Spring Web MVC*.
[C] Explain any two methods of *JdbcTemplate* class with proper example.
[D] Write a *Pointcut expression* for below Description in AOP:
 - Access specifier: *private*.
 - Return type of method: *String*.
 - Method which start by character 'c' and it is having two parameters of *any type*.
 - Apply only for the method of '*Employee*' class available in '*org.java*' package.
[E] Explain *Around Advice* with proper example.
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